Applied Industrial Data Science

Week 5, Lecture 5: Baseline Modelling on Validated Data

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Six words for today

Term	What it means	For example
Baseline modelling	Build the simplest credible model first, and score it	The bar every later model must clear
Naive baseline	A model that copies a number you already have	Order what sold yesterday
Heuristic model	A rule of thumb whose numbers a person chose	Same day last week, plus 10%
Regression	Fitted: finds the numbers linking inputs to output	A promotion adds 48 loaves
Feature	An input the model is allowed to use	Day of the week; a promotion
Evaluation metric	The rule that turns all the misses into one number	Average loaves out per day

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Baselines: start with the simplest guess

Word	In plain words	You already do it
Naive baseline	Copy a number you already have	"It will take as long as yesterday"
Baseline modelling	Score that guess first; it becomes the bar	Does an app beat yesterday's time?

A baseline is not meant to be good.
It is meant to be beaten.

If something clever cannot beat a copy of yesterday, it is not clever enough to use.

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Two kinds of rule: chosen and fitted

Word	In plain words	You already do it
Heuristic model	A rule of thumb; you chose the number	"Leave 15 minutes early if it rains"
Regression	The same rule; the data chose the number	A month of trips says rain adds 11

Same kind of rule.
The difference is who chose the number.

A heuristic comes from experience. A regression comes from data, so you can check it against the data.

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What goes in, and how you score it

Word	In plain words	You already do it
Feature	An input you know before deciding	Is it raining? Is it Friday?
Evaluation metric	One number for how wrong a rule was	Average minutes off, over a month

A feature must be known before you decide.
A metric scores every rule the same way.

That is all six words. Now the same six in a bakery, where a wrong guess costs bread or customers.

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WEEK 5

17:00. Tomorrow's bread order is due. The manager already has a rule.

Your model has to beat it.

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Three verdicts, and what they compare

01 Beats the rule
Lower error than the incumbent, on the same days, same metric.

02 Matches the rule
No better, so not worth the cost of changing anything.

03 Loses to the rule
Worse than what the manager does already. Back to the notebook.

A verdict is a comparison, never a score on its own.

First the data, then four ways to write down tomorrow's number. One of them is the manager's.

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Two weeks of bread, one store

Day	Last week	This week
Monday	120	122
Tuesday	118	121
Wednesday	125	119
Thursday	130	134
Friday	160	210
Saturday	190	185
Sunday	95	98

Write down Friday's order for next week. What number, and where did it come from?

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Four ways to write down a number

Method	The rule	Friday's prediction
Naive	Whatever sold yesterday	134, Thursday's figure
Mean	Average of last week	134
Seasonal naive	Same day last week	160
Manager's rule	Same day last week, plus 10%, rounded up	176
What actually sold		210

A model is any rule that turns what you know into a number you act on. A naive baseline is the laziest one: it copies a number you already have.

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Scoring one rule by hand

Day	Predicted, then sold	Miss, loaves
Monday	120, then 122	2
Tuesday	118, then 121	3
Wednesday	125, then 119	6
Thursday	130, then 134	4
Friday	160, then 210	50
Saturday	190, then 185	5
Sunday	95, then 98	3

Seven misses add to 73. Divided by 7: about 10 loaves a day. That is mean absolute error, and that is all it is.

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Score them all the same way

Method	Average miss, loaves	Worst day
Naive	33 (27+1+2+15+76+25+87 = 233, ÷ 7 = 33.3)	Sunday, 87 too many
Mean	29 (12+13+15+0+76+51+36 = 203, ÷ 7 = 29.0)	Friday, 76 short
Manager's rule	16 (10+9+19+9+34+24+7 = 112, ÷ 7 = 16.0)	Friday, 34 short
Seasonal naive	10 (2+3+6+4+50+5+3 = 73, ÷ 7 = 10.4)	Friday, 50 short

A metric is the rule that turns all the misses into one number, so two models can be compared. This one is mean absolute error, or MAE.

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THE FIRST FINDING

The rule had never been measured.

Not because it was bad. Because nobody thought of it as a model.

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The metric is a decision:

Method	Average miss	Days run short
Manager's rule	16	1 of 7
Seasonal naive	10	5 of 7

Change the metric and the ranking changes. Squared error flips it too.

Which error costs more, a loaf wasted or a customer with no bread? That answer picks the metric, and the metric picks the winner.

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Why squared error changes the winner

Score	Seasonal naive	Manager's rule
Average miss	10 loaves	16 loaves
Average squared miss	371 (2,599 ÷ 7)	343 (2,404 ÷ 7)
Friday, of the week's total	2,500 of 2,599	1,156 of 2,404

Squaring makes one big miss count more than many small ones.

A 50-loaf miss squares to 2,500, a 3-loaf miss to 9. Choose it when one bad day costs more than several slightly bad ones.

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The day every model missed

Method	Friday	Miss
Naive and mean	134	76 short
Manager's rule	176	34 short
Seasonal naive	160	50 short

A residual is the gap between what you predicted and what happened.

Different rules, different misses, one direction. Every one of them was short on Friday. That is not noise.

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What none of them knew

Question	Answer	Where it came from
What was different on Friday?	A promotion	The manager, in ten seconds
Could a model have known?	Yes, in advance	The promotions calendar
Is it a one-off?	No, every month	The last six months of promotions

A residual with a reason is a feature waiting to be added.

A feature is an input the model is allowed to use. Ask what it could not see, and whether you could have seen it in time.

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What a regression actually is

Part	Your journey	What it means
Starting number	25 minutes	Predicted when the input is zero: a dry day
Coefficient	11 per rainy day	How much the prediction moves per unit of input
Prediction	25 + 11 = 36 on a rainy day	Starting number plus coefficient times input

Prediction = starting number + coefficient x input

Both numbers are chosen by fitting, and both can be read out as a sentence. The starting number is also called the intercept.

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Regression as a baseline you can read

Part	FreshMart	What it means
Starting number	Same day last week	Start from what worked best
Coefficient	48 x promotion	A promotion adds about 48 loaves
Prediction	Friday: 160 + 48 = 208	Sold 210. Two loaves out.

Predicted loaves = same day last week + 48 x promotion

The starting number here is last week's sales, and fitting found one coefficient: 48 loaves for a promotion.

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Where the 48 came from

Past promotion	Same day last week, then sold	Extra loaves
March, a Friday	158, then 202	+44
April, a Saturday	188, then 239	+51
May, a Thursday	128, then 174	+46
June, a Friday	162, then 215	+53
July, a Wednesday	121, then 166	+45
August, a Friday	159, then 208	+49

Least squares, the usual way of fitting, chooses the average extra: 288 divided by 6 is 48.

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Score it again, same days, same metric

Method	Average miss, loaves	Days run short
Naive	33	3 of 7
Mean	29	2 of 7
Manager's rule	16	1 of 7
Seasonal naive	10	5 of 7
Seasonal + promotion	4	5 of 7

It beats the rule by twelve loaves a day. That sentence heads the interpretation note. Worst day is now Wednesday, six too many.

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A feature has to exist at five o'clock

Candidate feature	Known at 17:00 the day before?	Verdict
Day of the week	Yes, obviously	Use it
Promotion tomorrow	Yes, planned a week ahead	Use it
Weather forecast	Yes, but it might be wrong	Use with care
Today's sales so far	Yes, up to 17:00	Use it
Tomorrow's footfall	No, it has not happened	Never
Tomorrow's sales	No, that is the answer	Never

The test for every feature is the same: at the moment the decision is made, is this number known?

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Rules are models too, and sometimes the right one

Situation	The rule	Keep it or replace it?
Rule loses by 12 loaves a day	Last week + 10%	Replace, if 12 loaves pays for the change
Rule loses by 2 loaves a day	Last week + 10%	Keep it. The margin is not worth a rollout
No history to fit on	Any sensible rule	Keep it. A rule is what you use before data

A heuristic is a rule of thumb: a model whose numbers a person chose, not a fitting procedure.

Keep the rule when the model wins by less than the cost of changing. Replacing a rule is a project, not a decision.

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Why the baseline is the most important model

Someone says	Without a baseline	With one
It is 95% accurate	Sounds excellent	The rule was 94%. So what?
The error is 4 loaves	Sounds small	Compared with 16. It is good.
We improved it	Improved from what?	From 10 to 4, on the same days
We should deploy it	On what evidence?	It beats the rule by 12 a day

Every claim about a model is a comparison. The baseline is the thing it is compared with.

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EXIT REFLECTION

Compared with what?

Every number you report this term has to be able to answer that in one sentence.

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In the lab: the plant, not the bakery

Question	FreshMart	Your plant file
What is predicted?	Tomorrow's loaves	Next hour's energy, per line
The incumbent rule?	Last week + 10%	Same as last hour
The residual to explain?	Friday's promotion	The 112 degree hour

Same method, a different target, a week of your own plant data.

The file has no verified failures, so you do not predict failures. You predict next hour's energy, and you beat last hour's.

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In the lab, you measure the rule first

IN THE LAB

Score the incumbent rule before you build anything
Three baselines, by hand, then by code
One regression with features that exist in time
Find the column in your file that leaks
Write the interpretation note

WEEK 6

Interpretation and uncertainty
What the coefficient means,
and whether four loaves is good enough

Bring a model that beats the rule, and the metric on which it does not.

The second one is what the note is for.

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